Auditable Product-Stable Window Selection for Strong-Motion Records

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Abstract

Strong-motion products depend on the processing interval used to compute peak motion, waveform energy, and duration-sensitive measures. Fixed windows are simple to deploy in batch workflows. Their product retention can change sharply across archives. We evaluate this behavior with 53,463 three-component records from InstanceGM and K-NET. Each candidate window is audited by three checks: retention of full-record peak ground acceleration, retention of relative waveform energy, and inclusion of the full-record peak time. Fixed 42.00 s windows show strong dataset dependence. On InstanceGM, feature-onset, energy-onset, and catalog-P fixed windows have instability rates of 81.25%, 69.91%, and 73.03%. On K-NET, the same windows have instability rates of 4.15%, 3.10%, and 8.25%. We then evaluate a waveform-derived selector that chooses the shortest candidate passing the product checks and assigns the full record only when every shorter candidate fails. Only 0.84% of records are assigned to full-record processing, with median selected-window durations of 84.94 s for

InstanceGM and 24.66 s for K-NET. The selector recovers 25,468 InstanceGM feature-onset fixed-window failures and 21,913 InstanceGM energy-onset fixed-window failures under the same audit. A 5% damping response-spectrum audit gives overall PSA-retention failure rates of 12.98%, 22.26%, and 32.28% for feature-onset fixed windows at 0.2 s, 1.0 s, and 3.0 s. The shortest-stable selector reduces the same rates to 0.02%, 0.87%, and 5.56% (9, 465, and 2,972 records). Threshold tests make the operating trade-off explicit: the 0.95 energy-retention criterion keeps full-record assignment rare, whereas a 0.98 criterion raises the overall assignment rate to 46.84%. The results define a reproducible offline windowing audit for strong-motion product preparation.

Plain Language Summary

Strong-motion records need processing windows that keep the shaking used by engineering and seismological products. A fixed 42.00 s window can lose useful motion in one archive and include more time than needed in another. We test windows by checking whether they keep the full-record peak motion, enough waveform energy, and the time of the full-record peak. A simple offline selector chooses the shortest candidate window that passes these checks. The selector gives longer typical windows for InstanceGM and shorter typical windows for K-NET, with rare full-record assignment at the stated thresholds.

Introduction

Strong-motion records support earthquake engineering, shaking assessment, ground-motion model development, and post-event review. Products such as peak ground acceleration, energy measures, response spectra, and duration-related metrics depend on the part of the record used during processing. Archive-scale pipelines often use fixed-duration windows because they are easy to implement and easy to document. A single fixed interval can miss late product-relevant motion in one archive and retain unnecessary waveform in another.

These dependencies are a central issue in strong-motion processing and record-quality work (Douglas, 2003; Boore and Bommer, 2005; Boore et al., 2012).

Operational systems and services already cover many surrounding parts of the strong-motion processing problem. PRISM combines batch processing with a review interface for strong-motion records (Jones et al., 2017). The USGS gmprocess workflow supports automated near-real-time ground-motion processing (Thompson et al., 2025). European RRSM and ESM services provide rapid raw strong-motion parameters and reviewed engineering strong-motion products (Cauzzi et al., 2016; Luzi et al., 2016). Recent deep-learning work estimates P arrival time and high-pass corner frequency for strong-motion displacement processing (Inocente and Maruyama, 2026), and national product systems calculate PGA, PGV, PGD, intensity, duration, Fourier spectra, response spectra, and related outputs (Liu et al., 2025). This study isolates the record-level processing-window duration used to retain product-relevant motion before those products are reported.

Fixed windows are also common in public waveform datasets because they make records uniform, easy to index, and reusable across studies. That practical value creates a product-quality requirement: the retained interval needs to be checked against the strong-motion quantities computed from the record. The record-level question is whether the selected interval preserves peak motion, waveform energy, and spectral amplitudes used by downstream users. This study audits that question directly.

In a production strong-motion workflow, the window is a record-level processing decision made before product tables are exported. The same interval supports peak values, response spectra, energy-like measures, duration measures, and later review of flagged records. A useful selector keeps product-relevant motion, exposes the records that need full-record processing, and leaves an audit trail that can be checked after archive updates. Response spectra and duration measures are standard engineering products, which makes window retention a product-level requirement beyond plotting convenience (Trifunac and Brady, 1975; Dobry et al., 1978; Chopra, 2017).

This study treats processing-window selection as a product-retention problem. Arrival estimates help locate the beginning of shaking. The processing window also retains the quantities used downstream. The target output is an offline record-level window with traceable product checks. This framing matches strong-motion archive preparation and batch product generation, where the full record is available before a final processing window is stored.

We use InstanceGM and K-NET to test the same audit on archives with different product-window behavior. InstanceGM provides a large Italian strong-motion and seismic waveform archive through the INSTANCE data family (Michelini et al., 2021; see Data and Resources). K-NET provides Japanese nationwide strong-motion recordings operated by NIED (Aoi et al., 2004; Okada et al., 2004; see Data and Resources). Both datasets are evaluated at 100 Hz median sampling rate, keeping the comparison focused on waveform scale and window policy. Figure 1 summarizes the workflow.

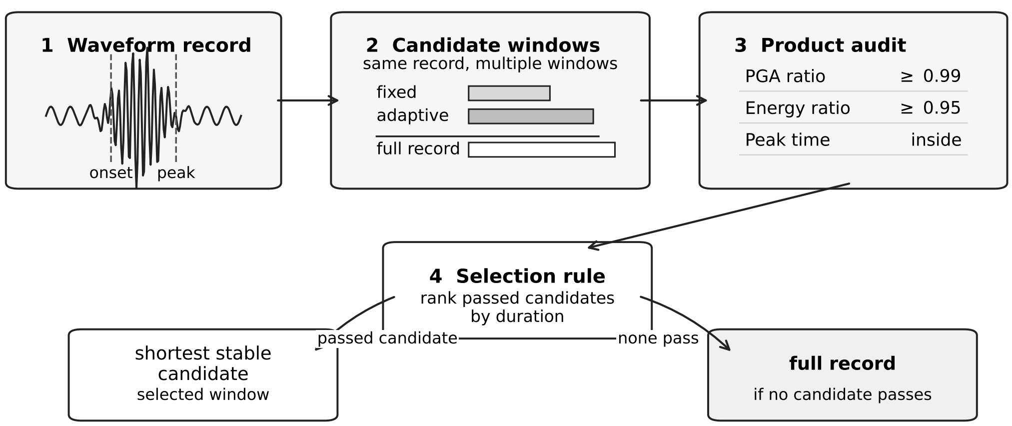

Figure 1: Workflow for offline product-stable window selection. Each waveform record is converted into candidate processing windows. Candidate windows are evaluated by retention of full-record PGA, relative energy, and inclusion of the full-record peak. The selector chooses the shortest stable non-full candidate and assigns the full record when no non-full candidate passes the product-retention checks.

The paper makes four contributions. First, it defines product stability for strong-motion processing windows using peak retention, energy retention, and peak-time inclusion. Second, it quantifies the dataset dependence of common fixed windows across 53,463 waveform records. Third, it evaluates a no-catalog, waveform-derived shortest-stable selector with explicit full-record accounting. Fourth, it reports product-impact recovery, threshold sensitivity, and response-spectrum retention as auditable engineering-product checks.

Data

The waveform audit contains 53,463 records with successful feature extraction (Table 1).

Table 1: Dataset summary for the 53,463-record waveform audit. Duration is median [p05-p95].

Dataset	Records	Events	Stations	Catalog P	Duration (s)	Median M
InstanceGM	31,344	9,927	568	31,344	120.00 [120.00-120.00]	3.4
K-NET	22,119	1,528	921	21,141	119.00 [60.00-120.00]	4.3

InstanceGM contributes 31,344 records from 9,927 events and 568 stations. K-NET contributes 22,119 records from 1,528 events and 921 stations. The two datasets have a median sampling rate of 100 Hz. InstanceGM records in the audit have a median duration of 120.00 s. K-NET records have a median duration of 119.00 s, with a 5th to 95th percentile range from 60 s to 120 s.

Records are grouped into low-magnitude background, M3-M4 small-event, and M4+ strong-motion strata (Table 2). InstanceGM contributes 9,000 low-magnitude background records, 9,000 M3-M4 records, and 13,344 M4+ records. K-NET contributes 21 low-magnitude background records, 5,647 M3-M4 records, 15,473 M4+ records, and 978 records with incomplete catalog timing for catalog-P comparison. The K-NET low-magnitude group is too small for a separate low-magnitude conclusion. It is retained in the full audit and reported as a data-availability note.

K-NET waveforms are read from the full converted archive and high-pass filtered at 1 Hz before feature extraction. This preprocessing matches the earlier K-NET waveform representation used in the project and separates raw acceleration offsets from product-window behavior. Strong-motion filtering and baseline choices can alter downstream amplitudes and spectra, so preprocessing is kept explicit (Boore, 2001; Douglas and Boore, 2011). Catalog

Table 2: Priority-stratum summary used in the waveform audit.

Dataset	Stratum	Records	Median duration (s)	Median M
InstanceGM	Low magnitude	9,000	120.00	2.2
InstanceGM	M3-M4	9,000	120.00	3.2
InstanceGM	M4+	13,344	120.00	4.4
K-NET	Low magnitude	21	60.00	2.9
K-NET	M3-M4	5,647	119.00	3.8
K-NET	M4+	15,473	119.00	4.5
K-NET	Other timing	978	119.00	4.4

P times support comparator windows and diagnostics. The main selector uses waveform-derived candidates.

Methods

Waveform Features and Candidate Windows

For each three-component record, we form a vector-amplitude signal from the standardized component traces. The waveform-derived feature set contains an effective onset, an energy onset, an energy end, and significant duration. These features define five candidate window families:

1. Feature-onset fixed: from 2 s before effective onset to 40 s after effective onset, yielding a 42.00 s processing interval.
2. Energy-onset fixed: from 2 s before energy onset to 40 s after energy onset.
3. Catalog-P fixed: from 2 s before catalog P to 40 s after catalog P. This is an arrival-time comparator.
4. Adaptive energy-end: from 2 s before effective onset to 3 s after waveform energy end.
5. Full record: the complete record, used as the upper-bound processing interval.

The main selector considers feature-onset fixed, energy-onset fixed, and adaptive energy-end candidates. Catalog-P fixed windows remain in the audit as arrival-time-based comparators.

Product-Stability Criteria

Each candidate is evaluated against the full record. Let PGA_{full} and E_{full} denote the full-record peak vector amplitude and vector-signal energy. Let PGA_{win} and E_{win} denote the same quantities inside a candidate window. Let $t_{\text{peak,full}}$ denote the time of the full-record peak. A candidate is stable when all checks pass:

$$\frac{PGA_{\text{win}}}{PGA_{\text{full}}} \geq 0.99, \quad \frac{E_{\text{win}}}{E_{\text{full}}} \geq 0.95, \quad t_{\text{peak,full}} \in w.$$

The energy term follows the Arias-style idea of integrating squared motion over time, applied here as a relative vector-signal energy criterion (Arias, 1970). The threshold sensitivity analysis repeats the selector with alternative energy-retention criteria.

Shortest-Stable Selector

For each record, the selector filters candidate windows by the three product-stability checks, ranks the stable non-full candidates by duration, and chooses the shortest one. If the filter yields no stable non-full candidate, the selector assigns the full record. In compact form,

$$w^* = \arg\min_{w \in C_{\text{record}}} \text{duration}(w) \quad \text{subject to} \quad \text{stable}(w) = 1.$$

The selected window passes the stated audit by construction. The performance information comes from fixed-window failure rates, selected duration, candidate usage, full-record assignment, and product-impact changes relative to fixed-window baselines.

Response-Spectrum Retention Audit

We add an independent response-spectrum retention audit to connect processing windows with engineering strong-motion products. For each selected or fixed window, we compute 5% damping pseudo-spectral acceleration (PSA) at 0.2 s, 1.0 s, and 3.0 s and compare it with the full-record PSA (Chopra, 2017). A window has a PSA-retention failure when

$$\frac{PSA_{\mathrm{win}}(T)}{PSA_{\mathrm{full}}(T)} < 0.95.$$

This audit is separate from the selector. The selector uses PGA retention, relative energy retention, and peak-time inclusion; the response-spectrum check evaluates whether those selected windows also preserve period-dependent motion measures.

Results

Fixed Windows Are Dataset-Dependent

Fixed 42.00 s windows have different product-retention behavior in the two datasets (Figure 2; Table 3). InstanceGM fixed-window instability is high: 81.25% for feature-onset fixed windows, 69.91% for energy-onset fixed windows, and 73.03% for catalog-P fixed windows. K-NET fixed-window instability is much lower: 4.15%, 3.10%, and 8.25% for the same three windows. Under this audit, the same fixed duration acts as a dataset-dependent product-window policy.

The adaptive energy-end candidate reduces instability to 1.45% for InstanceGM and 0.23% for K-NET. It also exposes the scale difference between datasets: the median adaptive duration is 84.12 s for InstanceGM and 24.66 s for K-NET.

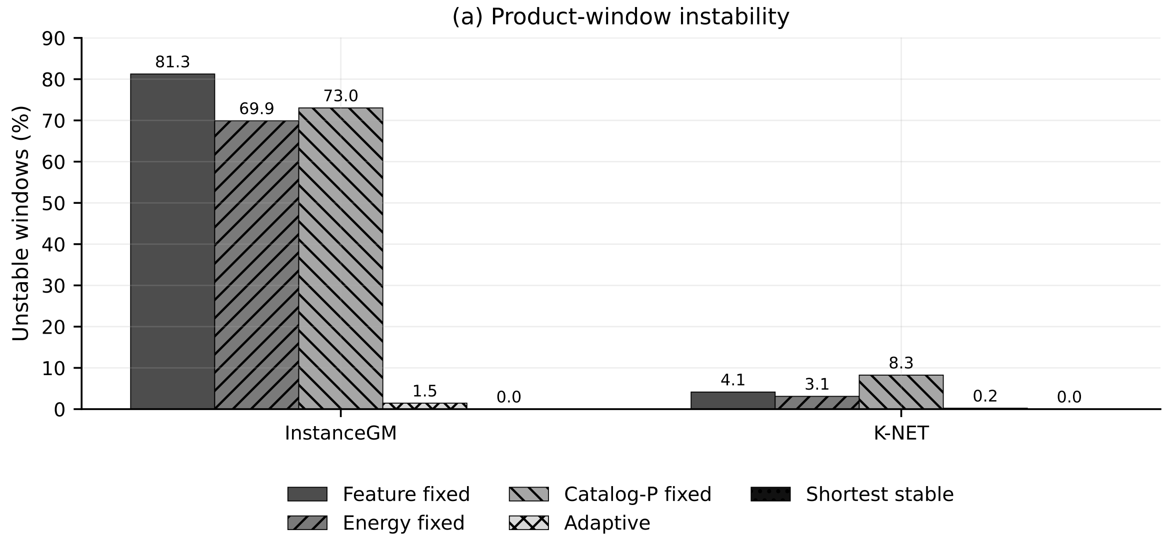

Figure 2: Product-window instability for fixed, adaptive, and selected windows on InstanceGM and K-NET records. Fixed 42.00 s windows fail frequently on InstanceGM and less often on K-NET, indicating dataset-dependent product retention. The shortest-stable selector removes product-retention failures under the stated audit because it filters candidates by the same criteria.

Table 3: Product-window stability summary for fixed, adaptive, and selected windows in the ALL priority group.

Dataset	Policy	Unstable (%)	Median window (s)	p05 energy	Full record (%)
InstanceGM	Feature fixed	81.25	42.00	0.080	0.00
InstanceGM	Energy fixed	69.91	42.00	0.228	0.00
InstanceGM	Catalog-P fixed	73.03	42.00	0.190	0.00
InstanceGM	Adaptive	1.45	84.12	0.953	0.00
InstanceGM	Shortest stable	0.00	84.94	0.954	1.31
K-NET	Feature fixed	4.15	42.00	0.957	0.00
K-NET	Energy fixed	3.10	42.00	0.963	0.00
K-NET	Catalog-P fixed	8.25	42.00	0.958	0.00
K-NET	Adaptive	0.23	24.66	0.960	0.00
K-NET	Shortest stable	0.00	24.66	0.960	0.18

Magnitude Strata Preserve the Dataset Contrast

The M3-M4 strata show the same dataset contrast (Table 3). In InstanceGM M3-M4 records, feature-onset and energy-onset fixed windows have instability rates of 77.19% and 61.01%. The selector assigns 1.79% of records to the full interval and has a median selected duration of 71.14 s in the same stratum. In K-NET M3-M4 records, feature-onset and energy-onset fixed windows have instability rates of 1.63% and 1.61%. The selector assigns 0.30% of records to the full interval and has a median selected duration of 19.39 s. The windowing issue follows archive and waveform-scale behavior. Magnitude strata provide an internal check on that pattern.

Shortest-Stable Selection Preserves Products with Limited Full-Record Assignment

The shortest-stable selector assigns 0.84% of records to full-record processing under the default product-stability criteria (Figure 3; Table 3). Dataset-level full-record assignment is 1.31% for InstanceGM and 0.18% for K-NET. The full audit contains 451 records assigned to the full interval, including 412 from InstanceGM and 39 from K-NET. The median selected duration is 84.94 s for InstanceGM and 24.66 s for K-NET. The selected windows pass the three product-retention checks by design; assignment rate and duration summarize the operating cost.

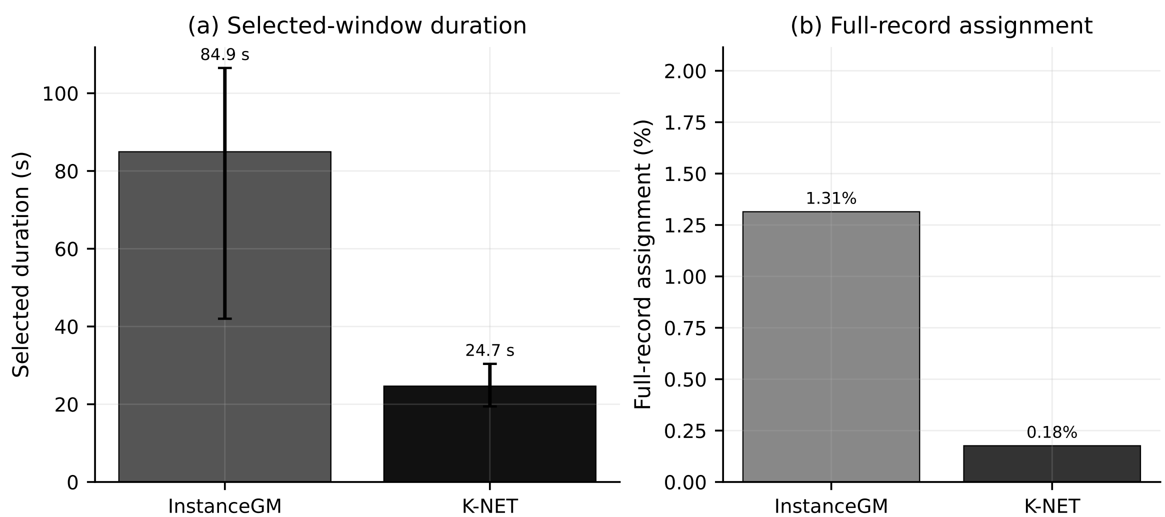

Figure 3: Selected-window duration and full-record assignment rate for the shortest-stable selector. The selector chooses longer typical windows for InstanceGM and shorter typical windows for K-NET. Only a small fraction of records are assigned to the full interval under the default PGA-retention and energy-retention criteria.

Candidate usage follows the dataset-scale difference. On InstanceGM, the selector chooses the adaptive energy-end candidate for 84.86% of records and the energy-onset fixed candidate for 13.52%. On K-NET, it chooses the adaptive energy-end candidate for 97.26%

and the energy-onset fixed candidate for 2.44%. Full-record assignment occurs for 0.84% of all records.

Product Impact Relative to Fixed Baselines

The selector changes product retention and selected-window length relative to fixed baselines (Figure 4). On InstanceGM, it resolves 25,468 feature-onset fixed-window failures, 21,913 energy-onset fixed-window failures, and 22,889 catalog-P fixed-window failures under the product-retention audit. The median energy-retention gain relative to feature-onset fixed windows is 0.347, and the median duration change is 42.94 s. These values indicate that many InstanceGM fixed windows end before sufficient product-relevant energy is retained.

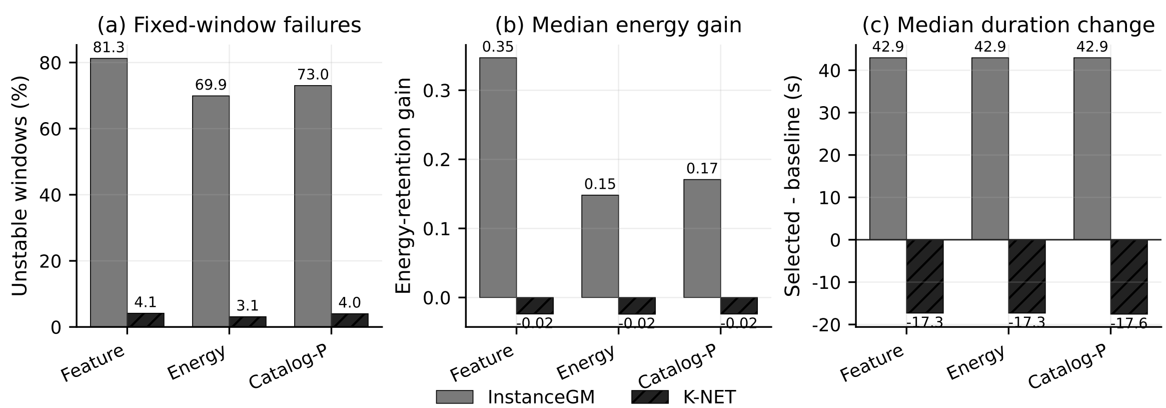

Figure 4: Product impact relative to fixed-window baselines. The panels show fixed-window instability, median energy-retention gain, and selected-minus-baseline duration change for feature-onset, energy-onset, and catalog-P fixed windows.

K-NET has fewer fixed-window failures. The selector resolves 917 feature-onset fixed-window failures, 686 energy-onset fixed-window failures, and 847 catalog-P fixed-window failures. The median duration change relative to feature-onset fixed windows is -17.31 s. The adaptive energy-end candidate often passes the product-retention checks with a shorter window than the 42.00 s fixed baseline.

Threshold Sensitivity

The selector is sensitive to the energy-retention criterion (Figure 5). With $PGA \geq 0.99$ and $E \geq 0.95$, the overall full-record assignment rate is 0.84%. Tightening energy retention to 0.98 raises the rate to 46.84% overall. InstanceGM full-record assignment increases from 1.31% at 0.95 energy retention to 71.84% at 0.98. K-NET assignment increases from 0.18% to 11.42%. The sensitivity analysis makes this trade-off explicit: stricter energy retention yields a more conservative selector and substantially more full-record assignments.

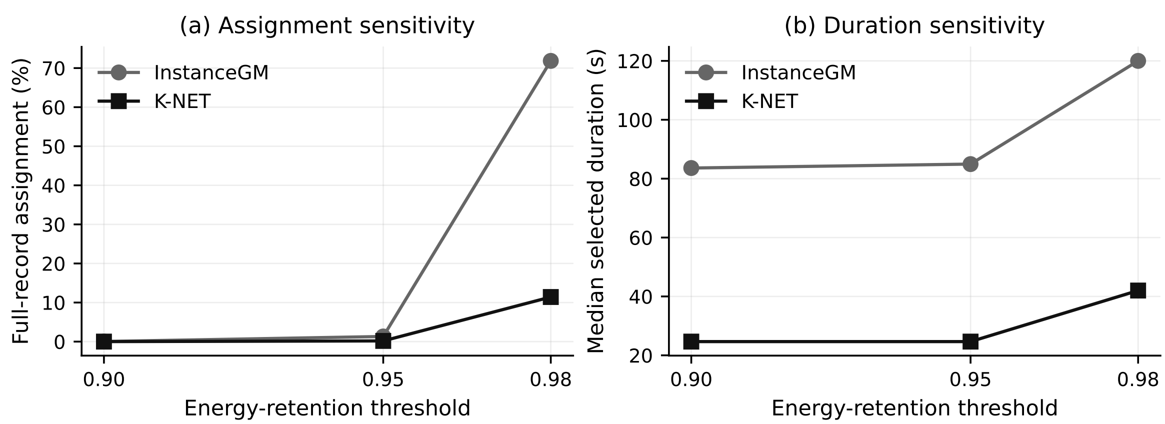

Figure 5: Sensitivity of the shortest-stable selector to the energy-retention criterion with the PGA-retention criterion fixed at 0.99. The default 0.95 energy-retention criterion keeps full-record assignment rare. A stricter 0.98 energy-retention criterion substantially increases full-record assignment, especially for InstanceGM.

Response-Spectrum Retention

The response-spectrum audit strengthens the product-window interpretation (Figure 6; Table 4). Across all 53,463 records, feature-onset fixed windows have PSA-retention failure rates of 12.98%, 22.26%, and 32.28% at 0.2 s, 1.0 s, and 3.0 s. The shortest-stable selector reduces the same rates to 0.02%, 0.87%, and 5.56%, corresponding to 9, 465, and 2,972 records.

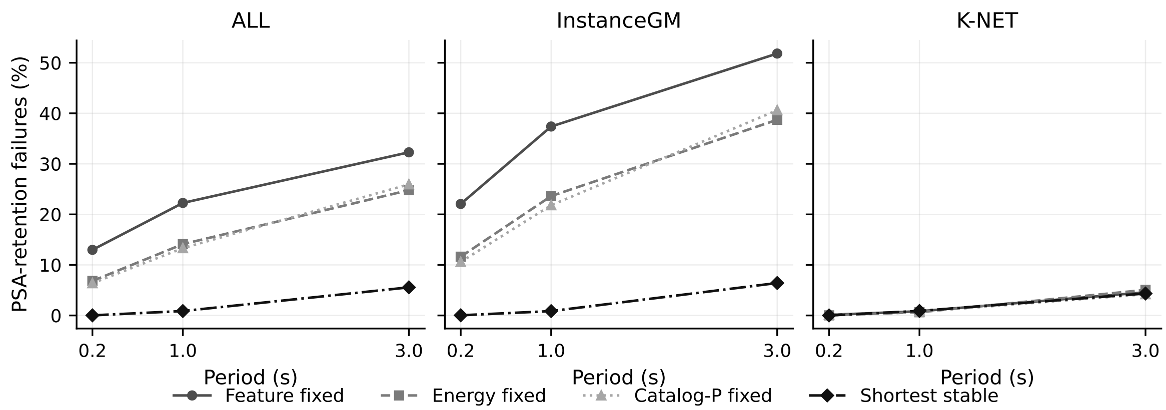

Figure 6: Response-spectrum retention at 5% damping. Panels compare PSA-retention failure rates at 0.2 s, 1.0 s, and 3.0 s for fixed windows and the shortest-stable selector. The selected windows reduce overall PSA-retention failures from 12.98%, 22.26%, and 32.28% for feature-onset fixed windows to 0.02%, 0.87%, and 5.56%.

Table 4: Response-spectrum retention summary. Values are 5% damping PSA-retention failure percentages relative to the full record.

Dataset	Policy	0.2 s failure (%)	1.0 s failure (%)	3.0 s failure (%)
ALL	Feature fixed	12.98	22.26	32.28
ALL	Shortest stable	0.02	0.87	5.56
InstanceGM	Feature fixed	22.06	37.39	51.83
InstanceGM	Shortest stable	0.03	0.86	6.42
K-NET	Feature fixed	0.11	0.83	4.59
K-NET	Shortest stable	0.00	0.89	4.34

The 3.0 s period exposes the main residual risk. On InstanceGM, feature-onset fixed windows have 51.83% PSA-retention failures at 3.0 s, and the selector has 6.42%. On K-NET, the same comparison is 4.59% and 4.34%. The selector substantially improves long-period retention for InstanceGM without increasing K-NET losses. This result clarifies the engineering relevance of the spectrum audit: short-period and intermediate-period products are largely preserved under the selected windows, while long-period PSA remains the product most likely to expose residual windowing loss. Records failing the 3.0 s PSA threshold are listed in the record-level audit material for long-period structural-response or soft-site product review.

Discussion

The results show that strong-motion processing windows need product-retention checks together with selected-window length. InstanceGM and K-NET show distinct window scales under a common audit. A fixed 42.00 s window is too short for many InstanceGM records and longer than needed for many K-NET records. The product-check selector adapts to that difference, and its operating cost is summarized by full-record assignment and selected duration.

The dominant failure mechanism is energy truncation. In InstanceGM, all 25,468 feature-onset fixed-window failures have energy-retention loss under the audit, and 8,504 also lose PGA. In K-NET, 916 of 917 feature-onset fixed-window failures have energy-retention loss, and 25 have PGA loss. The same 42.00 s interval interacts with archive-specific waveform duration and tail energy. This mechanism explains why a single fixed duration transfers poorly between archives even when both datasets are sampled at 100 Hz and processed by the same product-retention criteria.

The method leaves a clear audit trail. Each selected window can be traced to its candidate type, product-retention values, peak-inclusion status, and full-record assignment state. This record-level traceability matters for strong-motion product preparation because processing decisions need to survive later review and archive updates.

The record-level audit packet connects the aggregate results to practical review. It includes examples where fixed windows miss late product-relevant motion and the selector restores retention, examples where K-NET records keep the product checks with compact windows, and boundary cases assigned to the full record. These cases let an operator inspect why a record received its processing window without using hidden labels or manual arrival picks.

These results support an offline product-window policy. The full record is available before the processing window is finalized, and each selected window is evaluated against full-record products. The 0.95 energy-retention criterion is a design parameter supported by the sensitivity analysis. A stricter 0.98 criterion pushes many InstanceGM records to full-record processing. The response-spectrum audit adds an engineering-product check and shows that the selector retains most 0.2-3.0 s PSA values, with the largest residual loss at 3.0 s. The remaining 3.0 s failures identify records that need conservative handling for long-period spectral products. The selected windows still give a large product-level gain at 0.2 s, 1.0 s, and most 3.0 s records. Future work can start from this explicit product-stability baseline and test group-conditioned thresholds on an external validation archive.

Conclusions

We evaluated product-stable window selection for 53,463 strong-motion records from InstanceGM and K-NET. Fixed 42.00 s windows show strong dataset dependence, with high instability on InstanceGM and much lower instability on K-NET. The shortest-stable selector uses waveform-derived candidates and product-retention checks to choose compact processing windows. Under the default audit, full-record assignment is 0.84% overall, while median selected duration differs sharply by dataset: 84.94 s for InstanceGM and 24.66 s for K-NET. The response-spectrum audit shows that overall PSA-retention failures drop from 32.28% for feature-onset fixed windows to 5.56% for the selected windows at 3.0 s. The resulting contribution is an auditable offline product-window policy with explicit thresholds, full-record assignment accounting, cross-dataset sensitivity, and engineering-product retention checks.

Data and Resources

Waveforms from the InstanceGM/INSTANCE data family and K-NET were used in this study. K-NET waveforms were converted with explicit UD -> Z, NS -> N, and EW -> E component mapping, and K-NET waveform features were computed after 1 Hz high-pass preprocessing. The reproducibility release contains the source code, focused tests, manifest and worklist files, waveform-feature summaries, window-stability summaries, selector summaries, product-impact summaries, threshold-sensitivity summaries, response-spectrum audits, record-level audit cases, figure sources, checksums, and command log. The public release is archived at https://github.com/zhouhaoyiu/strong-motion-product-window-qc/releases/tag/v0.1.0. InstanceGM/INSTANCE data were accessed through https://doi.org/10.13127/INSTANCE on 16 June 2026. K-NET/NIED data were accessed through https://doi.org/10.17598/NIED.0004 on 16 June 2026. Raw waveform archives are not redistributed and remain subject to provider terms. Code and focused tests are released under the MIT License; derived summaries, figures, record-audit plots, manuscript-support metadata, and documentation are released under CC BY 4.0.

An AI-assisted language editing tool was used only to polish manuscript wording. The authors checked the scientific content, numerical outputs, source code, figure data, and final manuscript.

Acknowledgments

The authors thank the data providers of the InstanceGM/INSTANCE data family and the National Research Institute for Earth Science and Disaster Resilience (NIED) K-NET program for making waveform data available. This research received no external funding.

Declaration of Competing Interests

The authors declare no competing interests.

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